

# TERM PROJECT PROPOSAL

*Advanced Business Statistics: Business Data Analytics Using R*

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|------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Course Title</b>    | Advanced Business Statistics: Business Data Analytics Using R Programming                           |
| <b>Project Title</b>   | Analyzing E-Commerce Sales Patterns in the Korean Market Using Exploratory Data Analysis (EDA) in R |
| <b>Team Members</b>    | List all team members with student IDs                                                              |
| <b>Submission Date</b> | _____                                                                                               |
| <b>Class Code</b>      | IBT4109 – Group 001                                                                                 |

## 1. Title of the Project

Analyzing E-Commerce Sales Patterns in the Korean Market Using Exploratory Data Analysis (EDA) in R

## 2. Introduction and Project Overview

South Korea is one of the most digitally advanced e-commerce markets in the world, consistently ranking among the top five globally in terms of e-commerce penetration and per-capita online spending. With a highly connected population, dominant mobile commerce culture, and powerful domestic platforms such as Coupang, Naver Shopping, and Gmarket, the Korean online retail market offers a uniquely rich environment for studying consumer purchasing behavior and sales dynamics.

Despite this prominence, the structural patterns driving e-commerce sales — including seasonal demand cycles, category-level trends, regional differences, and the impact of major promotional events (such as 11.11 sales or Chuseok shopping periods) — remain insufficiently analyzed using open data and reproducible statistical methods.

This project proposes a comprehensive Exploratory Data Analysis (EDA) of Korean e-commerce sales patterns using publicly available datasets from the Korean Statistical Information Service (KOSIS) and supplementary sources, implemented entirely in R. The project will apply the full EDA toolkit — distribution analysis, time-series visualization, correlation analysis, hypothesis testing, and graphical summaries — to uncover actionable patterns in how Korean consumers shop online. This aligns directly with Weeks 9–14 of the IBT4109 course syllabus, covering data preprocessing, subsetting, ANOVA, and graphical EDA methods.

### 3. Background, Significance, and Objectives

#### 3.1 Academic and Industry Background

The academic study of e-commerce sales patterns draws on a rich interdisciplinary body of literature combining consumer behavior theory, data mining, and business analytics. Time-series decomposition, seasonal adjustment, and statistical hypothesis testing are established methods in this domain, widely applied to retail datasets in the United States, China, and Europe. However, Korea-specific analyses using open data remain comparatively rare in the English-language academic literature, representing a meaningful research gap.

From an industry perspective, understanding sales patterns is critical for inventory planning, logistics resource allocation, marketing campaign timing, and platform monetization strategies. Korean e-commerce platforms process millions of transactions daily, and even marginal improvements in demand forecasting accuracy can translate into substantial cost savings.

#### 3.2 Significance and Research Gap

Existing Korean e-commerce studies often rely on proprietary transaction-level data accessible only to large firms or government agencies. This project addresses that gap by demonstrating that meaningful and statistically rigorous analysis can be performed using publicly available aggregate statistics — making the methodology accessible to researchers, small businesses, and policymakers alike.

#### 3.3 Research Objectives and Questions

The primary objectives of this project are:

- To identify and visualize seasonal and trend patterns in Korean e-commerce sales across major product categories.
- To statistically test whether sales levels differ significantly across seasons, product categories, and regions.
- To analyze the relationship between macroeconomic indicators (consumer confidence, unemployment rate) and online retail sales volume.
- To provide evidence-based insights for e-commerce businesses operating in the Korean market.

The guiding research questions are:

- Which product categories exhibit the strongest seasonal sales cycles in Korean e-commerce?
- Are there statistically significant differences in online sales volume across quarters (ANOVA)?
- How does consumer confidence index correlate with e-commerce transaction volume?
- What regional differences in e-commerce adoption and spending exist within Korea?

### 4. Analytical Method and Research Design

#### 4.1 Overall Approach

This project uses a secondary data analysis design with a descriptive and inferential statistical approach. Rather than building a predictive model, the emphasis is on thorough exploration, visualization, and hypothesis testing — making it a textbook EDA project aligned with the course methodology for Weeks 13 and 14.

## 4.2 Methodology

The analysis will proceed in three stages:

- Stage 1 – Descriptive EDA: Summary statistics, frequency distributions, histograms, boxplots, and time-series line charts by category and quarter.
- Stage 2 – Inferential Testing: One-way ANOVA to test for quarterly sales differences; post-hoc Tukey HSD tests to identify specific group differences; independent t-tests for pre/post event comparisons (e.g., before vs. after major promotional campaigns).
- Stage 3 – Correlation & Association Analysis: Pearson correlation matrices between sales, consumer confidence, unemployment rate, and CPI; scatter plots with regression lines using ggplot2.

## 4.3 Software Tools

All analyses will be performed in R (version 4.x) using the following packages:

- tidyverse (dplyr, ggplot2, tidyr): Core data manipulation and visualization
- readxl / readr: Importing KOSIS and Statista datasets
- lubridate: Parsing and handling date/time variables for time-series analysis
- ggplot2 + scales: Advanced visualization (faceted plots, seasonal decomposition charts)
- stats (base R): ANOVA, t-tests, Pearson correlation
- corrplot: Correlation heatmaps
- forecast / tseries: Optional time-series decomposition (STL method)

## 4.4 Justification

EDA is the most appropriate starting methodology for this research because the field lacks a strong a priori theory specifying which variables will be the dominant drivers of Korean e-commerce sales patterns. A thorough exploratory phase is necessary before any confirmatory modeling. Furthermore, ANOVA and t-tests are ideal for answering categorical group comparison questions (seasonal, regional, categorical), and these methods are explicitly covered in this course.

# 5. Data Description and Preprocessing

## 5.1 Data Sources

The following publicly accessible datasets will be used:

- KOSIS (Korean Statistical Information Service): Monthly online shopping transaction value by product category and payment method. Available at: [kosis.kr/eng](http://kosis.kr/eng)
- Statistics Korea (KOSTAT): Consumer Price Index, consumer confidence index, household income data. Available at: [kostat.go.kr](http://kostat.go.kr)

- Statista Korea E-Commerce Reports: Supplementary market size and platform share data. Available at: [statista.com](https://www.statista.com) (university access)
- Bank of Korea (BOK) Economic Statistics System: GDP, unemployment rate, retail sales indices. Available at: [ecos.bok.or.kr](https://ecos.bok.or.kr)
- Naver DataLab (Public API): Search volume trends for major product categories — useful proxy for consumer interest and demand signals. Available at: [datalab.naver.com](https://datalab.naver.com)

## 5.2 Key Variables

| Variable Type | Variable Name             | Description                                               |
|---------------|---------------------------|-----------------------------------------------------------|
| Primary (Y)   | Online Sales Volume (KRW) | Monthly e-commerce transaction value by product category  |
| Primary (Y)   | Transaction Count         | Number of online orders per month (where available)       |
| Grouping (X1) | Product Category          | Fashion, Electronics, Food & Beverage, Beauty, etc.       |
| Grouping (X2) | Quarter / Season          | Q1–Q4 categorization for ANOVA analysis                   |
| Temporal (X3) | Year-Month                | Time index for trend and seasonality analysis (2019–2024) |
| Macro (X4)    | Consumer Confidence Index | Bank of Korea monthly CCI score                           |
| Macro (X5)    | Unemployment Rate (%)     | KOSTAT monthly unemployment for correlation analysis      |
| Regional (X6) | Metropolitan Region       | Seoul, Gyeonggi, Busan, etc. (where data permits)         |

## 5.3 Data Preprocessing Steps

The following preprocessing steps will be conducted in R:

- Data import and merging: Load KOSIS CSV exports and BOK time-series data; join by year-month key using `dplyr::left_join()`.
- Date parsing: Convert character date strings to Date objects using `lubridate::ymd()` and extract year, month, and quarter as new columns.
- Missing value treatment: Visualize missingness using `naniar::vis_miss()`; apply forward-fill for isolated missing months in time-series data.
- Category standardization: Harmonize product category labels across datasets (KOSIS uses Korean labels requiring translation and remapping).
- Normalization: Convert nominal sales values to real terms using the CPI deflator from KOSTAT to allow valid year-over-year comparisons.
- Outlier flagging: Identify anomalous months (e.g., COVID-19 lockdowns in 2020) and add indicator flags rather than removing those observations.

## 6. Project Constraints and Limitations

- Language barrier: KOSIS data is primarily in Korean, requiring translation and careful label mapping before analysis.
- Data aggregation level: Publicly available e-commerce statistics are typically aggregated by category and month; individual transaction-level data is not publicly accessible.
- Platform-level data: Data from individual platforms (Coupang, Gmarket) is proprietary and cannot be directly accessed, limiting platform-specific analysis.
- Causality limitations: EDA cannot establish causal relationships; observed correlations between macro indicators and sales may reflect confounding factors.
- Scope of regional data: Sub-national breakdowns are available for some but not all variables, which may limit regional comparison depth.

### 7. Project Plan and Timeline

The project will be conducted over six weeks (Weeks 9–14) as illustrated in the Gantt chart below. Shaded cells represent the active work period for each task.

| Task                                     | Wk 9 | Wk 10 | Wk 11 | Wk 12 | Wk 13 | Wk 14 |
|------------------------------------------|------|-------|-------|-------|-------|-------|
| Literature Review & Hypothesis Formation |      |       |       |       |       |       |
| Data Collection (KOSIS, Statista, Naver) |      |       |       |       |       |       |
| Data Cleaning & Preprocessing in R       |      |       |       |       |       |       |
| EDA: Visualization & Pattern Analysis    |      |       |       |       |       |       |
| Statistical Testing (ANOVA, T-test)      |      |       |       |       |       |       |
| Report Writing & Slide Preparation       |      |       |       |       |       |       |

Team responsibilities: Data sourcing and Korean-language data translation (Member 1), R coding for EDA and statistical tests (Member 2 & 3), visualization design and interpretation (Member 1 & 2), report writing and slide preparation (all members, coordinated by Member 3).

### 8. Expected Results and Contributions

We expect the EDA to reveal clear and statistically significant seasonal patterns in Korean e-commerce, particularly around Q4 (November–December) due to year-end promotional events, and Q1 dips following the holiday season. Specifically, we anticipate:

- Fashion and electronics categories will exhibit the strongest seasonal variation, with ANOVA showing significant quarterly differences ( $p < 0.05$ ).
- Consumer confidence index will show a moderate positive correlation with total e-commerce volume (expected  $r > 0.5$ ).
- Food and daily necessities will show steadier, less volatile sales patterns compared to discretionary categories.

- A significant structural break in sales patterns around Q1–Q2 2020 will be identifiable, reflecting COVID-19-induced shifts to online purchasing.

The contributions of this study include:

- A reproducible, open R-based EDA framework for Korean e-commerce data that other researchers can adapt.
- The first English-language, publicly reproducible statistical analysis of KOSIS e-commerce panel data.
- Practical insights for domestic and foreign firms targeting the Korean online retail market.
- A foundation for follow-up predictive modeling (e.g., SARIMA forecasting) in future research.

## 9. References

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